

Research Statement — Anirban Bhaduri

1. Introduction

A derived category packages the homological data of a geometric or algebraic object into a single structure, and much of the geometry of a variety can be recovered from it. A natural question is then how *complicated* such a category is: how efficiently can the whole category be assembled from a single object?

Answers are recorded by numerical invariants such as the Rouquier dimension and the Orlov spectrum, which are notoriously resistant to computation. Outside a short list of examples their values are unknown. My dissertation computes these invariants for bounded derived categories of coherent sheaves on weighted projective lines and of modules over path algebras of acyclic quivers, identifying structural mechanisms that explain the answers.

Let $\mathcal{T}$ be a triangulated category. A *classical generator* is an object G from which every other object can be obtained using cones, shifts, direct sums and direct summands; G is a *strong generator* if finitely many such steps suffice, and the least number of steps required is the *generation time* of G . The set of all generation times is the *Orlov spectrum* of $\mathcal{T}$, and its infimum is the *Rouquier dimension*. A *semiorthogonal decomposition* of $\mathcal{T}$ is a sequence of full subcategories, generating $\mathcal{T}$, between which morphisms run in only one direction.

These invariants were introduced and developed by Bondal and Van den Bergh [7], Rouquier [29], Orlov [21] and Ballard–Favero–Katzarkov [3]. Explicit computations remain rare: Ballard–Favero–Katzarkov [3] determined the full Orlov spectrum for the derived category of a Dynkin quiver of type A_n and obtained partial information for projective space $\mathbb{P}^n$.

Example 1.1. Let $\mathbb{P}^1$ be the projective line over an algebraically closed field k ; the category $\mathrm{coh}(\mathbb{P}^1)$ is hereditary, and every object splits as a direct sum of a torsion and a torsion-free part. Orlov [21, Proposition 15] showed that any object of the form $\mathcal{O}(a)[m] \oplus \mathcal{O}(b)[n]$ with $a \neq b$ generates $D_{\mathrm{coh}}^b(\mathbb{P}^1)$ in a single cone, whereas an object of the form $\mathcal{O}(a)[m] \oplus \mathcal{O}_p[n]$, with p a closed point, requires two. Hence the Orlov spectrum of $D_{\mathrm{coh}}^b(\mathbb{P}^1)$ is $\{1, 2\}$ and its Rouquier dimension is 1 [29]. The category also admits the semiorthogonal decomposition $D_{\mathrm{coh}}^b(\mathbb{P}^1) \simeq \langle \mathcal{O}, \mathcal{O}(1) \rangle$.

2. Derived Categories of Quivers and Weighted Projective Line

My graduate research concerns path algebras of quivers, their bounded derived categories, and the relation of these to algebraic geometry. A quiver is a directed graph, and its paths form a k -basis of the path algebra kQ . When $D_{\mathrm{coh}}^b(X)$ admits a tilting object, the resulting equivalence $D_{\mathrm{coh}}^b(X) \simeq D_{\mathrm{mod}}^b(kQ)$ turns geometric questions into questions about finite-dimensional algebras; in Example 1.1, Q is the Kronecker quiver

$$1 \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} 2$$

Path algebras of acyclic quivers are hereditary, so Ext^i vanishes for $i \geq 2$; in Dynkin type there are moreover only finitely many indecomposables up to isomorphism, which makes these categories especially tractable. Recall that Ballard–Favero–Katzarkov [3] computed the Orlov spectrum of $D_{\mathrm{mod}}^b(kA_n)$ to be $\{0, 1, \dots, n-1\}$, where A_n is

$$1 \longrightarrow 2 \longrightarrow \cdots \longrightarrow n$$

2.1. Orlov Spectra of Weighted Projective Lines (*Status: Ongoing, draft available on request*)

Weighted projective lines $\mathbb{P}(a, b)$, when viewed as a stacky curve, forms a natural generalization of the projective line. It was first introduced by Geigle and Lenzen in [14] and since then, it has found relevance in the fields of algebraic geometry and representation theory. These can be found in the works of Krause–Chen in [11], Ruan–Wang in [30], Elagin in [12] and Otani in [23].

I study the derived category of weighted projective lines $\mathbb{P}(1, n)$, explicitly compute generation times, determine the Orlov spectrum. We have $D_{\text{coh}}^b(\mathbb{P}(1, n)) \cong D_{\text{mod}}^b(k\tilde{A}_{n+1})$, where $\tilde{A}_{n+1}$ is the extended Dynkin quiver

$$1 \xrightarrow{\quad} 2 \longrightarrow \cdots \longrightarrow n+1$$

The case of $\mathbb{P}(1, 2)$ was settled in my dissertation. In my subsequent joint work with Matthew Ballard, we have closed the general case.

Theorem 2.1 (Ballard, B.) *The Orlov spectrum of $D_{\text{coh}}^b(\mathbb{P}(1, n))$ is $\{1, 2, \dots, n+1\}$. Consequently, the Orlov spectrum of $D_{\text{mod}}^b(k\tilde{A}_{n+1})$ is $\{1, 2, \dots, n+1\}$.*

We show explicit generators with each generation time in the spectrum. We provide bounds by decomposing a category using its semiorthogonal decomposition and computing ghost ideals. Our work generalizes the results on Orlov spectra for the Dynkin quiver A_{n+1} in [3] and the projective line $\mathbb{P}^1$ in [21] to the extended case and weighted case, respectively.

2.2. Geometric Model for A_n and Summand Minimal Generator (*Status: Ongoing, draft available on request*)

A geometric model for $D_{\text{mod}}^b(kA_n)$ is given by a disk with $n+1$ boundary points marked $0, 1, \dots, n$ with arcs joining any two points representing an indecomposable in this category. Morphisms and extensions between two modules are determined by interaction between two arcs at the boundary or in the interior.

Geometric models for categories are an active research area as we can see from the works of Araya [1], Broomhead [8], Oppen–Plamondon–Schroll [20] and Gratz–Stevenson [15]. In my ongoing collaborative project with Ballard, we have shown that

Theorem 2.2 (Ballard, B.) *Let $G = \oplus_{i=1}^m G_i$ be an object in $D_{\text{mod}}^b(kA_n)$ and for each i , G_i is an indecomposable, and let α_i be the corresponding arc of G_i in the geometric model. G is a strong generator if and only if (a) $\cup_i \alpha_i$ is a connected topological set, and (b) each boundary point is an endpoint of arc α_i for some i .*

Using geometric model, we explore a new class of generators in triangulated categories.

Definition 2.3. Let $\mathcal{T}$ be a Krull-Schmidt triangulated category and let $G = \oplus_i G_i$ be a strong generator with each G_i indecomposable. G is a summand minimal generator if no proper summand of G generates the category.

One of our goals is to explore the summand minimal generators of $D_{\text{mod}}^b(kA_n)$ using its geometric model. We show that:

Theorem 2.4 (Ballard, B.) *There exists a unique class of summand minimal generators in $D_{\text{mod}}^b(kA_n)$ of length $\lceil \frac{n+1}{2} \rceil$ and generation time one.*

Here, length denotes the number of indecomposable summands of these generators. It can be shown that there is a spectrum for length of summand minimal generators:

Theorem 2.5 (Ballard, B.) *The length spectrum for summand minimal generators of $D_{\text{mod}}^b(kA_n)$ is*

$$\left\{ \left\lceil \frac{n+1}{2} \right\rceil, \dots, n \right\}.$$

3. Derived Categories of Non-Commutative Schemes

Derived categories and non-commutative algebraic geometry have been a topic of interest in recent times due to the works of Van den Bergh, Špenko, Orlov, Wemyss, Lowen and others. Orlov [22] defines a smooth non-commutative scheme as the category of perfect complexes over a differential graded algebra, and Van den Bergh [31] introduced non-commutative crepant resolutions as an algebraic replacement for crepant resolutions of singularities.

3.1. Dimension Theory of Non-commutative Curves (*Status: Arxiv preprint*)

Reiten–Van den Bergh [28] gave a categorical notion of a non-commutative curve.

Definition 3.1. A *non-commutative curve* is a k -linear Ext-finite abelian category $\mathcal{A}$ with homological dimension one.

They further show [28, Theorem V.1.2] that $\mathcal{A}$ is either the category of modules over the path algebra of an acyclic quiver, or the category of coherent sheaves on a smooth projective stacky curve. This dichotomy organizes the computation of dimensions: the quiver side has been settled by Elagin [12] for the Rouquier, diagonal and Serre dimensions and by Qiu [26] for the global dimension, and the results below settle the stacky side.

Several invariants are in play. In earlier work with Lunts [13], Elagin introduced the Serre dimension; Ballard–Favero [2] defined the diagonal dimension and bounded it for varieties and Deligne–Mumford stacks; and Ikeda–Qiu [16] introduced the global dimension of stability conditions. In the commutative case these are known in low dimension: Orlov [21, Theorem 6] showed that the Rouquier dimension of a smooth projective curve of genus ≥ 1 is one, the diagonal dimension of a smooth projective curve was determined in [19], and the global dimension of an orbifold projective line in [23].

In collaboration with Antonios–Alexandros Robotis and Isaac Goldberg, we extend these results to the stacky setting and answer the following.

Question 3.2 *What are the values of the Rouquier, diagonal, Serre and global dimensions of a non-commutative curve?*

Theorem 3.3 (B., Goldberg, Robotis, [6]) *Let $\mathcal{X}$ be an orbifold curve. Then:*

- (a) *the Rouquier dimension of $D_{\text{coh}}^b(\mathcal{X})$ is one;*
- (b) *$\text{Ddim}(\mathcal{X}) = 1$ if $\deg(\omega_{\mathcal{X}}) < 0$, and $\text{Ddim}(\mathcal{X}) = 2$ otherwise;*
- (c) *if $\mathcal{X}$ is a smooth projective Deligne–Mumford stack of dimension n over a field k , then $\overline{\text{Sdim}}(D^b(\mathcal{X})) = \underline{\text{Sdim}}(D^b(\mathcal{X})) = n$;*
- (d) *if $\mathcal{X}$ is an orbifold curve over $\mathbb{C}$, then $\text{gldim}(\mathcal{X}) = 1$.*

Together with the computations on the quiver side, this determines the Rouquier, diagonal, Serre and global dimensions of every non-commutative curve.

3.2. Preservation for Generation (*Status: Published in Bulletin of the London Mathematical Society*)

In recent joint work with Lank and Dey [5], we studied non-commutative schemes $(X, \mathcal{A})$, where X is a scheme and $\mathcal{A}$ is a coherent $\mathcal{O}_X$ -algebra. Our goal was to relate generation

in such non-commutative settings to that of the underlying commutative scheme, and to establish bounds on Rouquier dimension. We proved the following theorem:

Theorem 3.4 (B., Dey, Lank, [5], Theorem 4.4) *Let X be a Noetherian J -2 scheme of finite Krull dimension, and let $\pi: \mathcal{O}_X \rightarrow \mathcal{A}$ be a coherent $\mathcal{O}_X$ -algebra with full support. If G is a classical generator of $D_{\text{coh}}^b(\mathcal{A})$, then $\mathbf{R}\pi_* G$ is a classical generator of $D_{\text{coh}}^b(X)$.*

This shows that generators on the non-commutative side descend to the commutative side. As a consequence, we obtained a general lower bound:

Theorem 3.5 (B., Dey, Lank, [5], Corollary 4.7) *If X is integral, Jacobson, catenary, Noetherian, and J -2, and if $\mathcal{A}$ is a coherent $\mathcal{O}_X$ -algebra with full support, then*

$$\text{Rdim}(D_{\text{coh}}^b(\mathcal{A})) \geq \dim(X).$$

These results connect the Rouquier dimension of non-commutative schemes to fundamental invariants of the underlying commutative variety, providing a categorical tool to measure their complexity.

3.3. Derived McKay Correspondence and Semi-Orthogonal Decomposition (*Status: Published in Advances in Mathematics*)

Similar to the notions of generation and dimension, the study of larger derived categories often requires breaking them down into more manageable pieces. A key tool for this is semiorthogonal decomposition, which allows us to understand a derived category by analyzing its smaller components that together generate the whole.

The Derived McKay Correspondence Conjecture states that for a finite subgroup $G \subset \text{SL}(n, \mathbb{C})$, the bounded derived category of coherent sheaves on any crepant resolution Y of $\mathbb{C}^n/G$ is equivalent to the derived category of finitely generated modules over the skew group algebra [17, 18]. In [4], a joint work originating from the AMS MRC 2023, we studied reflection groups of rank two generated by order-two reflections. Building on the conjecture of Polishchuk–Van den Bergh [24], which predicts that the components of the semi-orthogonal decomposition of $D^G(X)$ correspond to the irreducible representations of G , we proved the following:

Theorem 3.6 (B., Davidov, Faber, Honigs, McDonald, Overton-Walker, Spence, [4], Theorem A) *Let $G = G(2m, m, 2)$ with $m \geq 3$, or G_{12}, G_{13}, G_{22} . Then*

$$D^G(\mathbb{C}^2) \cong \langle D(B_1), \dots, D(B_r), E_1, \dots, E_n, D(\mathbb{C}^2/G) \rangle,$$

where B_i are the normalizations of the irreducible components of the branch divisor, E_j are exceptional objects, and $r + n + 1$ equals the number of irreducible representations of G .

Together with Potter’s work [25], this confirms the Polishchuk–Van den Bergh conjecture for groups $G \leq \text{GL}(2, \mathbb{C})$ generated by order-two reflections, via explicit computations on the H -Hilbert scheme.

4. Future Directions

4.1. Quivers and Weighted Projective Line

A natural generalization of Theorem 2.1 is the following.

Question 4.1 *What is the Orlov spectrum of $D^b(\text{mod-}kQ)$ for an acyclic quiver Q ?*

The proof of Theorem 2.1 derives the spectrum of $\tilde{A}_n$ from that of A_n , and this transfer is not specific to type A : given the spectrum in a Dynkin type, the same argument should yield the corresponding extended type. The finite types are therefore the bottleneck. Types

D and E admit no weighted projective line model, so I would compute them directly from the Auslander–Reiten quiver together with ghost ideal estimates, and then recover $\tilde{D}$ and $\tilde{E}$ by the transfer. A further direction is the 3-Kronecker quiver, whose spectrum would lead naturally to that of $D_{\text{coh}}^b(\mathbb{P}^2)$, for which Ballard–Favero–Katzarkov obtained only partial information. Another natural question that arises from my study of geometric model and I would want to explore is:

Question 4.2 *How can we visualize strong generation and ghost sequences in $D_{\text{coh}}^b \mathbb{P}(a, b)$ via a geometric model?*

This is a reasonable goal for the following reasons: (a) a geometric model for $D_{\text{coh}}^b \mathbb{P}(a, b)$ is explored in recent work like [10], and (b) generation in Dynkin quivers using a geometric model is one of my current ongoing projects. In the geometric model for kA_n , a summand minimal generator can be represented with a collection of arcs on a disc. A class of generators with similar orientation and same generation time can be obtained by action of a dihedral group D_{n+1} on such objects. I would be interested in exploring questions like:

Question 4.3 *Can we classify summand minimal generators in $D_{\text{mod}}^b(kA_n)$ modulo action by a dihedral group D_{n+1} ?*

I have made explicit computations and progress for specific cases in this direction. As a concrete example, I have shown that there are only two types of summand minimal generators of length three in $D_{\text{mod}}^b(kA_4)$, G_1 of generation time one and G_2 of generation time two, up to an action by D_5 . The rotations in D_{n+1} in these categories can be represented by the Auslander-Reiten translation functor, and reflections can be represented by the duality functor. I expect to make substantial progress on these questions within one year.

4.2. Non-Commutative Geometry

Following Orlov, a smooth non-commutative scheme is the category of perfect complexes over a differential graded algebra A . A recurring difficulty is the relationship between A and its zeroth homology algebra $H^0(A)$: it is known that $H^0(A)$ need not be smooth even when A is, so the passage between the derived and classical worlds destroys information associated with smoothness. The Works of Raedschelders and Stevenson [27] and of Brown, Soto Levins and Sridhar [9] suggests that coconnective algebras, those whose positive homology vanishes, are the right class in which to look. Silting objects are a "good" source of such algebras.

Question 4.4 *For which coconnective differential graded algebras A is $H^0(A)$ smooth, and how does generation in A compare with generation in $H^0(A)$?*

My joint work with Dey and Lank [5] bounds the Rouquier dimension of $D_{\text{coh}}^b(\mathcal{A})$ below by $\dim(X)$ by pushing classical generators forward along the structure morphism. The same strategy, transporting a generator along a morphism and controlling what the transport costs, applies to the map $A \rightarrow H^0(A)$, and this is where I intend to begin. I have been interested in these questions for several years and regard this as my principal long-term project.

Two further directions are at an earlier stage. The natural sequel to my work on non-commutative curves is the surface case; a classification of abelian categories of homological dimension two appears out of reach, so I would approach it through non-commutative resolutions of surface singularities and through 2-Calabi–Yau categories, where the relevant categories are explicit. Separately, silting objects govern the bounded t -structures on $D^b(\Lambda)$ with algebraic heart, and so connect generation to the geometry of stability manifolds. A

possible direction worth exploring would be to bring the arc models of Section 4.1 to bear. I expect both directions to develop through collaboration.

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